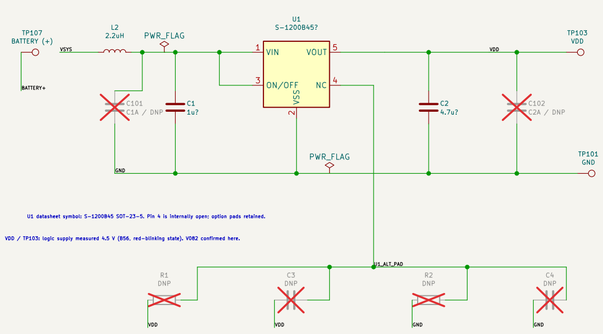
01 POWER / INPUT FILTER

H01: likely S-1200845 LDO. Nominal 4.5 V is a candidate, not a measured rail.



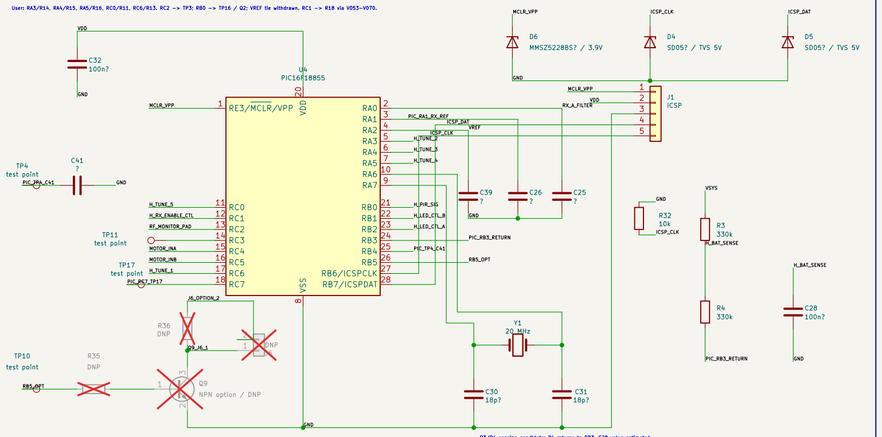
U1 datasheet symbol: S-1200B45 SOT-23-5. Pin 4 is internally open; option pads retained.

VDD / TP103: logic supply measured 4.5 V (B56, red-blinking state). VDB2 confirmed here.

Empty options: R1/C3 free ends join regulated VDD; R2/C4 free ends join GND.

02 PIC CONTROLLER / CLOCK / ICSP

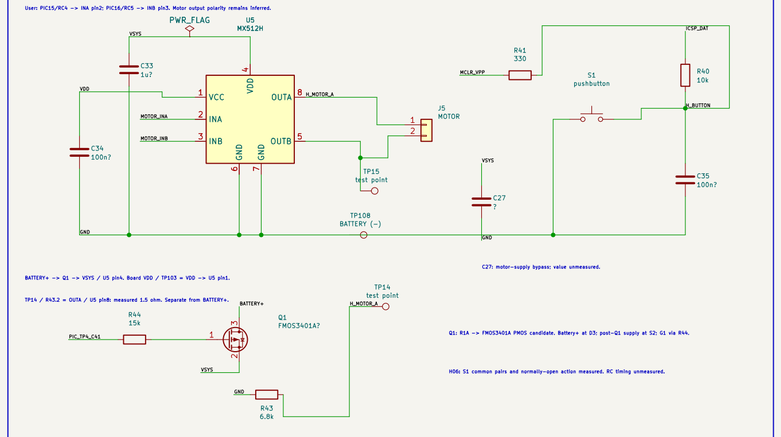
User: RA3/R14, RA4/R15, RA5/R16, RC0/R11, RC6/R13, RC2 -> TP3; RB0 -> TP16 / Q2; VREF tie withdrawn, RC1 -> R18 via V053-V070.



R3/R4 sensing candidate: R4 returns to RB3. C28 value estimated.

03 MOTOR / BUTTON / BATTERY INTERFACE

User: PIC15/RC4 -> INA pin2; PIC16/RC5 -> INB pin3. Motor output polarity remains inferred.



BATTERY+ -> Q1 -> VSYS / U5 pin4. Board VDD / TP103 -> VDD -> U5 pin1.

TP14 / R43.2 -> OUTA / U5 pin8: measured 1.5 ohm. Separate from BATTERY+.

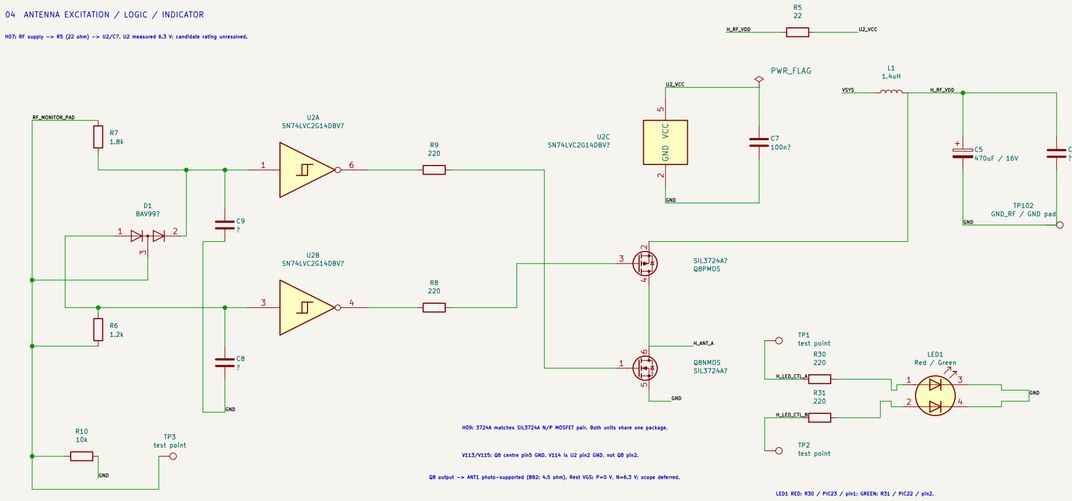
C27: motor-supply bypass: value unmeasured.

Q1: R1A -> FMOS3401A PMOS candidate. Battery+ at D3; post-Q1 supply at S2; G1 via R44.

H08: S1 common pairs and normally-open action measured. RC timing unmeasured.

04 ANTENNA EXCITATION / LOGIC / INDICATOR

H07: RF supply -> R5 (22 ohm) -> U2/C7. U2 measured 6.3 V: candidate rating unresolved.



H09: 3724A matches SIL3724A N/P MOSFET pair. Both units share one package.

V113/V115: Q8 centre pin5 GND. V114 is U2 pin2 GND, not Q8 pin2.

Q8 output -> ANT1 photo-supported (B82: 4.5 ohm). Rest VGS: P=0 V, N=6.3 V; scope deferred.

LED1 RED: R30 / PIC25 / pin1; GREEN: R31 / PIC22 / pin2.

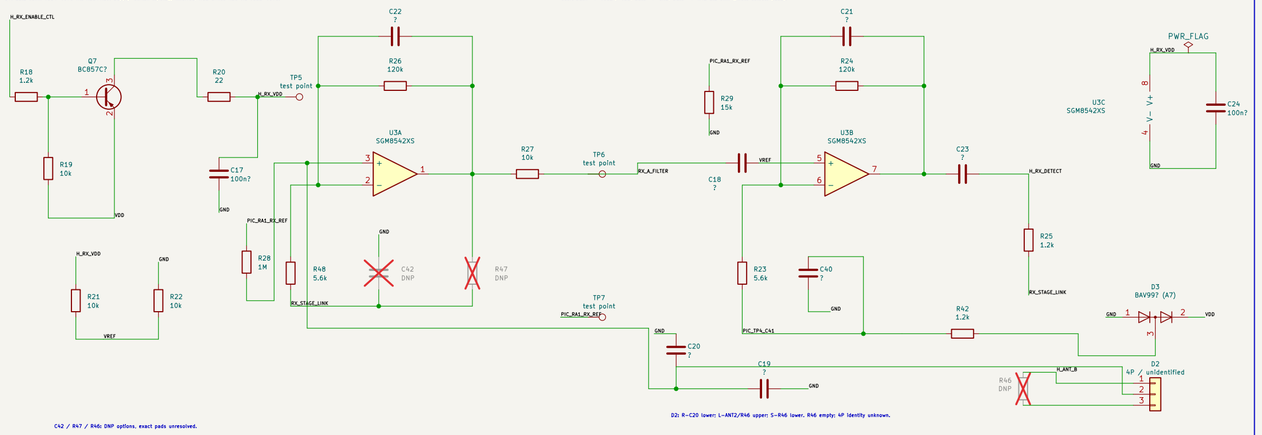
IMG_2431: TP3/PIC13 drives R6/R7/D1 midpoint. Diode identity and timing unverified.

05 RF RECEIVE / GAIN / ENVELOPE

H10–H12: working topology around photographed 120k feedback resistors. Filters, bias, and interstage routes inferred.

Q7: 11.2 kohm V064–V070 ffs R18+R19. Supply-switch topology inferred: RC1 drives R18 via V053–V070.

User: D3.R -> V082 / VDD; D3.L -> GND: D3.S -> R42 1.2 ohm (photo selects pad2).



C42 / R47 / R46: DNP options, exact pads unresolved.

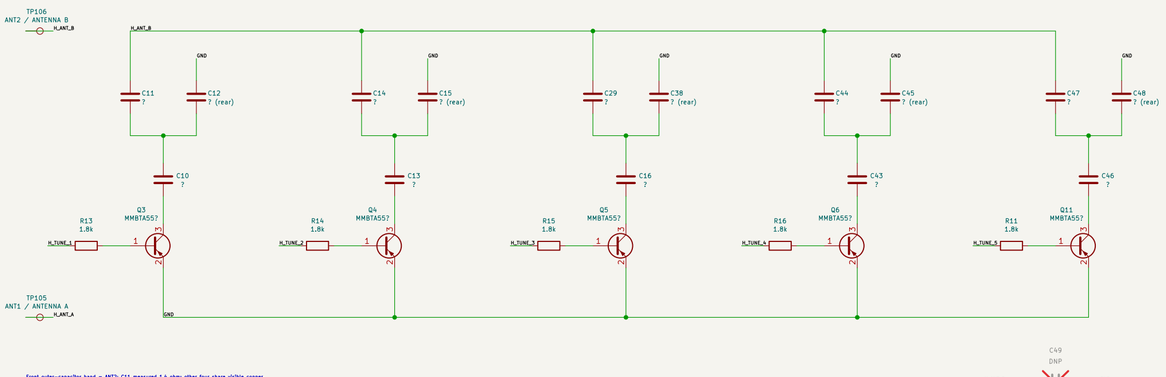
U3 alternative: MCP6002-/SN: same pin roles, electrical suitability to verify.

D2: R-C20 lower: L-ANT2/R46 upper: S-R46 lower. R46 empty; 4P identity unknown.

User: U3 pin5 / V079 / V080 = VREF (1 ohm): TP6 remains separate across C18.

06 SWITCHED ANTENNA CAPACITANCE

H13: 2H devices are probably PNP BJTs (MMBTA55 family; PBSS5140T is an alternative). Capacitor values unknown.



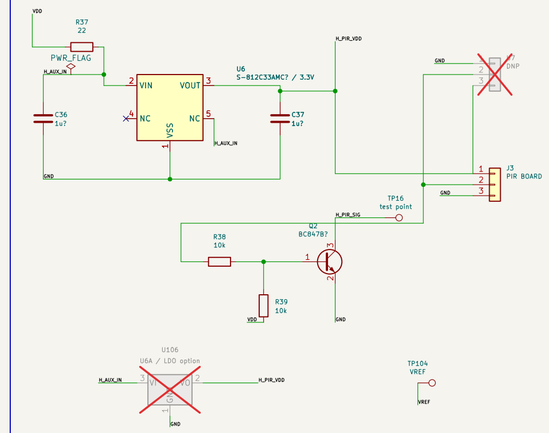
Front outer-capacitor bend = ANT2: C11 measured 1.4 ohm; other four share visible copper.

Rear capacitor returns remain GND. Front and rear capacitors are not drawn in parallel.

Q3 diode pattern disputes PNP pinout; 2H family and RF switching role remain unverified.

07 PIR / AUXILIARY RAIL

H14–H15: C2N supports S-812C33AMC 3.3V LDO. C/R3 ground supported by resistance.



U6: pin5 joins VIN (1 ohm); pin4 unused. PIR supply measured 3.3 V with sensor attached.

U6A: DNP alternative regulator: stock symbol does not identify the original part.

READING THIS WORKING RECONSTRUCTION

One sheet. Functional wiring is proposed: this is not a verified production circuit.

148 retained catalog entries: unsupported TP9 and R33 entries withdrawn.

H_* = inferred inter-block net. Other net names do not imply measurement.

? on a part value or pin function means an estimate or candidate identity.

Library pin numbers are mapped to photo pads in evidence/pin_crosswalk.json.

U2 and U3 use multiple symbol units on this same sheet, including supply units.

28 original local photo fragments retained: Q1.L–R43.2 withdrawn by user.

RA0 reaches TP6. TP11/TP17 local routes retained; hidden branches unproved.

DNP = unpopulated. An open pin means unresolved, not a certified no-connect.

PIR daughterboard internals are not yet traced; J3 represents its interface.

No capacitor value or rail voltage in this drawing was electrically measured.

See docs/RECONSTRUCTION.md for evidence, alternatives, and calculations.

v0.9.41 | KiCad 10.0.5 | 2026-10-15

TP11/TP17 retained; hidden branches and device roles remain qualified.

KiCad 10 libraries + four datasheet symbols | MCP authoring

Sheet: /

File: PetSafe_1001339.kicad_sch

Title: PetSafe 100–1339 R03 A | single-sheet reconstruction

Size: A2	Date: 2026-10-15	**Rev: 0.9.41**
KiCad E.D.A. 10.0.5		Id: 1/1